SMART INDIA HACKATHON 2025

TITLE PAGE

- Problem Statement ID - 25240
- Problem Statement Title - AI/ML based text-to-text machine translation from Nepalese and Sinhalese to English
- Theme - Heritage & Culture
- PS Category - Software
- Team ID -
- Team Name (Registered on portal)



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AI/ML based text-to-text machine translation from Nepalese and Sinhalese to English

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PROBLEM STATEMENT ANALYSIS



Literature inaccessible due to language barriers



Manual translation too slow and impractical

WHAT IS NEEDED?

Needed: Parallel Dataset, OCR, powerful hardware, integrated translation software

Outcome: All-in-one interactive tool for device discovery, vulnerability reports, exploits, and fixes

CORE FOCUS



Optical Character Recognition



Machine Translation

PROPOSED SOLUTION





This tool accepts images and PDFs as input



Automatically extracts all Nepali or Sinhalese text from these files



Translate the extracted text into English

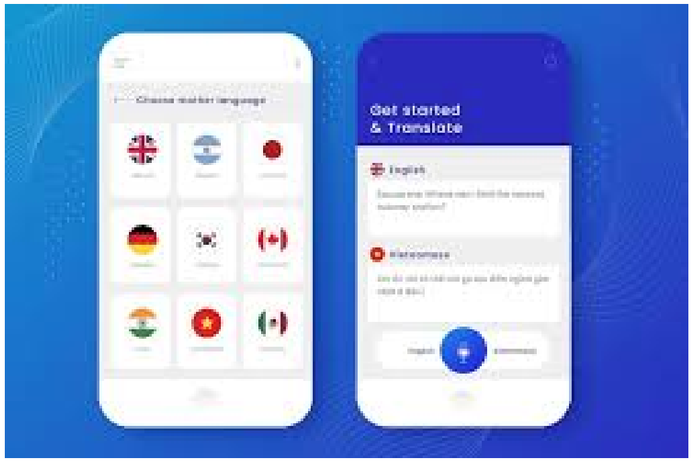
UNIQUENESS

Operates offline without needing an internet connection





Integrates a review interface for experts to validate and improve translations



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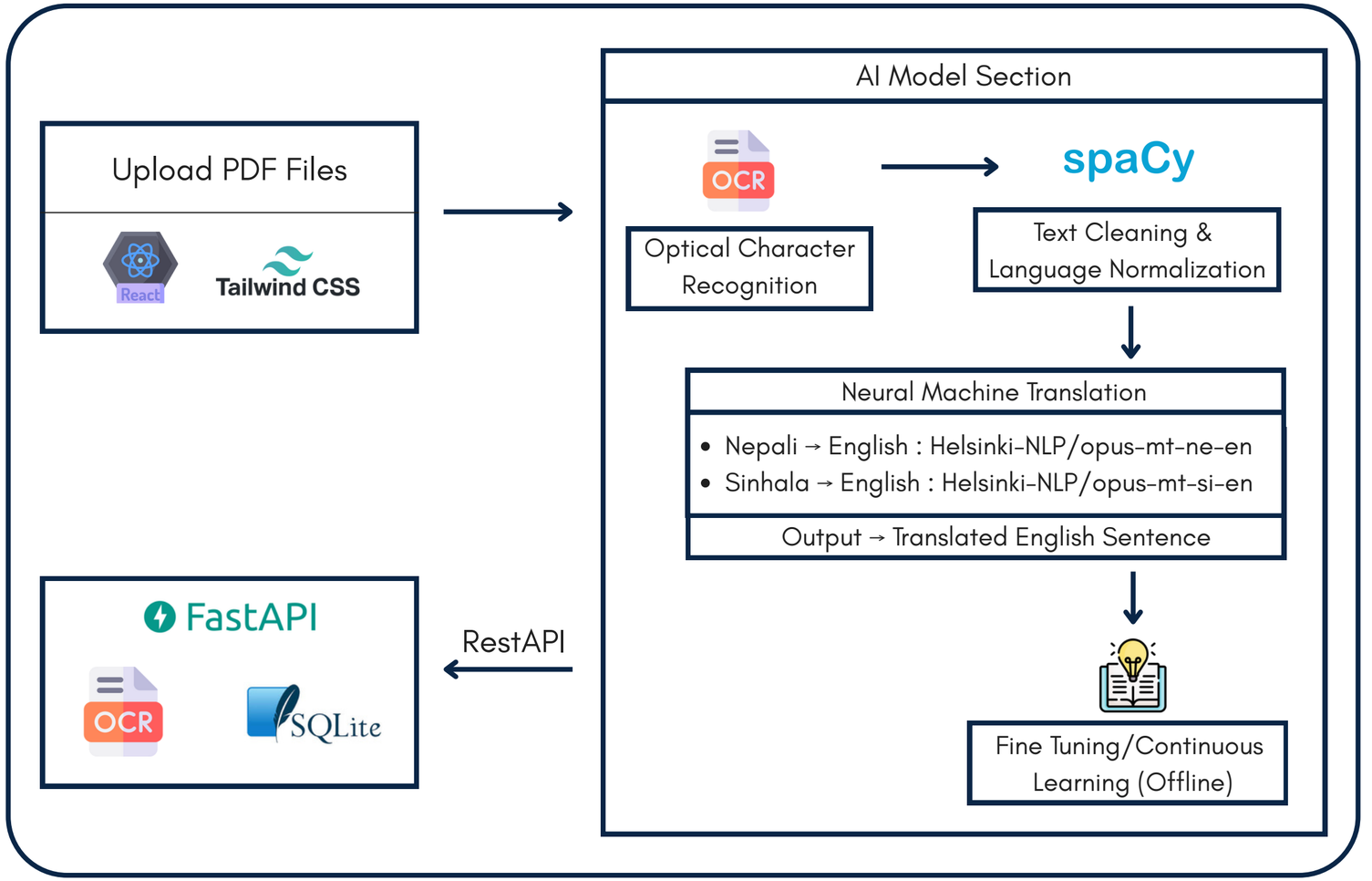
TECHNICAL APPROACH

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TECH STACK

Frontend	React, Tailwind CSS
Backend	FastAPI
OCR Engine	OpenCV, Tesseract
Language Detection	UNICODE
Fine-Tuning Module	PyTorch
Translation Engine	Hugging Face
NLP Preprocessing	spaCy
Feedback DB	SQLite
Deployment	docker

WORKFLOW



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FEASIBILITY AND VIABILITY

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Technical Feasibility

The model is feasible using existing OCR, NLP, and translation frameworks like Tesseract and mBART.



Operational Feasibility

Easily deployable as an offline desktop or web app for seamless translation operations.



Economic Feasibility

Cost-effective since it uses open-source tools and requires minimal hardware.

POTENTIAL CHALLENEGES AND RISKS



Data and System Challenges

Limited annotated datasets and script-specific OCR accuracy issues.



Human and Organizational Challenges

Need for skilled personnel to manage model training and validation.



Safety and Security Challenges

Ensuring data privacy, offline security, and preventing unauthorized access to translated documents.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

Data and Technology Strategies

Utilize multilingual datasets, fine-tune transformer models, and enhance the accuracy of optical character recognition (OCR).





Human and Organizational Strategies

Empowers staff by providing comprehensive training in AI workflows, and implement a robust review and feedback system to ensure continuous improvement and collaboration.

Safety and Security Strategies

To enhance safety and security, implement data encryption, ensure offline processing, and establish strict access control measures.

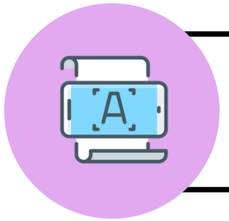


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IMPACT AND BENEFITS

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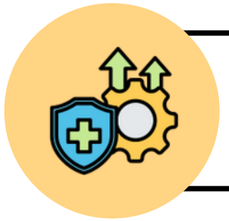
IMPACT



25% enhancement in digital preservation of heritage texts.



70% reduction in manual translation effort through automation.



60% faster document processing using AI-driven OCR and translation.



40% improvement in translation consistency compared to human-only efforts.

BENEFITS



The software will enable instant translation of Nepali and Sinhalese texts, significantly reducing time and effort compared to manual translation.



By bridging language barriers, the software promotes inclusivity, allowing non-native speakers to understand, enjoy, and interact with diverse cultural content.



It will make regional literature accessible globally, promoting cultural exchange, knowledge sharing, and appreciation of Nepalese and Sinhalese heritage.



Automation of translation tasks will increase productivity for organizations, freeing human translators to focus on more creative or complex language work.



The AI/ML-based solution will improve translation accuracy over time through continuous learning, reducing human errors and inconsistencies in text conversion.



Researchers, students, and linguists will gain easy access to a wide range of texts, enhancing educational resources and academic research opportunities.

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RESEARCH AND REFERENCES

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9. AI TranslateDocs - Translate Documents Online with AI. AI-driven document translation with OCR and layout preservation. Link here
10. Google Cloud Vision API - OCR Language Support. Detects and extracts text from images, supporting Nepali and Sinhala languages. Link here